SCC Branch Book

Structural Counterfactual Capacity, Minimal Carrier, and Endpoint Admissibility

Elements of Nested Fibrational Cosmology

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Front Block

Imported spine results.

(1) Book I: Observational Quotient Theorem (Thm. I.6.1); Collapse Gates CG-1–CG-6 (SRs I.1.4–I.1.9); Admissible-Class Canonicity (Thm. I.3.6); Behavioral Congruence Canonicity (Lem. I.3.5).
(2) Book II: Boundary Stabilization Theorem (Thm. II.8.1); Phase-A/B/C classification (Def. II.5.1); PASS as Derived Necessity (Thm. II.5.14); Binary Rigidity (Thm. II.5.8).
(3) Book III: Unified Coercive-Transport Inequality (Thm. III.5.1); Depth-Sum Convergence (Thm. III.5.4).
(4) Book IV: Universal Source Theorem (Thm. IV.7.1); five source-side rungs ORA–UC (Thms. IV.2.1–IV.3.1).
(5) Book V: Branch Projection Theorem (Prop. V.3.1); UBLT (Thm. V.7.4); Certified Branch Legitimacy Theorem (Thm. V.8.1).
(6) Book VII: Canonical Closure Schema Theorem (Thm. VII.6.4); No-Hidden-Upgrade Rule (Prop. VII.6.x); Non-Retroactivity (Thm. VII.6.y).

Branch-specific data.

(1) Declared SCC target category: the weak structural witness category $\mathcal{C}_{\mathrm{SCC}}$, whose objects are SCC quadruples (A_X, W_X, T_X, D_X) (support skeleton, witness family, transported invariant, defect ledger).
(2) The SCC terminal predicate τ_{SCC}: the structural counterfactual capacity condition on an SCC object.
(3) The SCC dependency spine: ORA $\Rightarrow$ CTM $\Rightarrow$ TIN $\Rightarrow$ MCS $\Rightarrow$ WeakGlue $\Rightarrow$ UC $\Rightarrow$ TSI.
(4) Four open obligations: O-SCC.MCS (main frontier), O-SCC.WeakGlue (prerequisite to MCS), O-SCC.UC, O-SCC.TSI.

Endpoint target. Persistence-selected structural counterfactual capacity: the certified structural property of an NFC-source-descended object that supports lawful counterfactual reasoning about its own observer-relative distinguished states, under the constraints of the canonical source image. No phenomenological, conscious, or experiential content is asserted.

Bridge stack.

(1) Licensed source-descended realization via F_{SCC} (ORA–UC rungs from Book IV).
(2) SCC Minimal Structural Carrier Selection (O-SCC.MCS): the main open burden.
(3) SCC Branch-Local Pairwise Amalgamation (O-SCC.WeakGlue): prerequisite to MCS discharge.
(4) Branchwise SCC completion $\mathcal{U}_{\mathrm{SCC}}$ (O-SCC.UC).
(5) SCC Target/Structural Identification (O-SCC.TSI): full endpoint admissibility beyond structural-label form.

Current status. **CERT-PROJ at most. Main frontier: O-SCC.MCS.**

The SCC branch is a lawful NFC descendant. All four core obligations (MCS, WeakGlue, UC, TSI) are open in canonical papers. The branch is at CERT-PROJ —

licensed projection at stated structural scope — and cannot advance to CERT-CLOSE until the MCS theorem is canonically proved.

SCC.0 Purpose

This branch studies the structural counterfactual capacity of NFC-source-descended objects. Its task: determine whether the canonical spine produces a certified object that supports lawful counterfactual reasoning about its own observer-relative distinguished states, under the no-smuggling discipline.

The SCC branch occupies a unique position in the NFC program. Its endpoint is the furthest from direct physical measurement and the most philosophically sensitive. The program's response to this is not to retreat from the endpoint but to impose the strictest possible labeling discipline: the endpoint label "structural counterfactual capacity" is the strongest admissible label consistent with the spine's theorems, and no phenomenological or ontological upgrade is licensed.

The canonical discipline has one specific consequence: the word "consciousness" and its cognates are not licensed at any stage in the canonical theorem flow. This is not a philosophical stance on whether consciousness is real; it is a status-honesty claim that the current canonical framework has not proved a consciousness theorem and does not use consciousness as a premise.

1. Branch Governance Preamble

Standing Rule 1.1 ([D]). (SCC Anti-Smuggling.) No target-side label for the SCC branch may introduce content beyond the certified source image. In particular: no phenomenology, no independent ontology, no private experiential primitive, no consciousness claim, and no import from empirical cognitive science may enter the canonical theorem flow without explicit declaration and separate canonical proof.

Standing Rule 1.2 ([D]). (Consciousness Non-Claim.) No theorem in this book licenses the label "consciousness," "subjective experience," "qualia," "phenomenal states," or any equivalent label. The strongest admissible endpoint label is *persistence-selected structural counterfactual capacity*. Any stronger label requires a separate canonical proof not present in this book.

Proposition 1.3 ([U]). (Formation.) *The SCC branch candidate* $\mathfrak{B}_{\mathrm{SCC}}$ *is constitutionally well-formed: all five components* $(\mathbf{U}, \mathcal{C}_{\mathrm{SCC}}, F_{\mathrm{SCC}}, \mathrm{Cert}_{\mathrm{SCC}}, \tau_{\mathrm{SCC}}, \mathrm{FF}_{\mathrm{SCC}})$ *are declared.*

Proof. All five components are declared in the front block and defined in Sections 3–5. By the Certified Branch Legitimacy Theorem (Book V, Thm. V.8.1), the branch is constitutionally formed. □

Proposition 1.4 ([U]). (Interpretive Label Admissibility.) *The label "persistence-selected structural counterfactual capacity" passes all four tests of the No-Hidden-Upgrade Rule (Book VII, Prop. VII.6.x): (i) it is eliminable (reduces to the structural predicate* τ_{SCC}*); (ii) it asserts no independent ontology; (iii) it imports no phenomenology; (iv) it does not depend on data outside the certified source image.*

Proof. Test (i): the label is eliminable in favor of the structural predicate on the SCC quadruple (A_X, W_X, T_X, D_X). Test (ii): SCC is a structural property of NFC configurations; no independent mental ontology is asserted. Test (iii): no experiential or phenomenal content enters the definition of τ_{SCC}. Test (iv): the predicate is computed entirely from the certified source image $F_{\mathrm{SCC}}(\mathbf{U})$. □

Theorem 1.5 ([U]). (SCC TSI Firewall.) *A proposed endpoint label L is admissible for the SCC branch if and only if none of the four interpretive failures occur:*

(i) *FI-IADD: L adds theorem content not in the certified spine image.*
(ii) *FI-IONT: L asserts an independent ontology.*
(iii) *FI-IPHEN: L imports phenomenology absent from the theorem content.*
(iv) *FI-ISRC: L depends on data outside the certified source image.*

The label "persistence-selected structural counterfactual capacity" passes all four tests. "Consciousness" and its cognates fail FI-IONT and FI-IPHEN at the current canonical stage.

Proof. The SCC TSI firewall is the No-Hidden-Upgrade Rule (Book VII, Prop. VII.6.x) applied to the SCC endpoint. Tests (i)–(iv) are the exact four tests of that rule. For the admissible label: Proposition 1.4. For consciousness: at the current stage, no theorem in the canonical spine or this branch book identifies the SCC structural predicate with phenomenal consciousness; importing that identification would constitute FI-IONT (independent mental ontology) and FI-IPHEN (phenomenology not in theorem content). □

Remark 1.6 ([D]). (Status vocabulary.) Standard FI-code and positive status alphabet applies. Current status: CERT-PROJ at most. No claim in this book exceeds CERT-PROJ.

2. Imported Spine Structure

Imports from Book I. The observational quotient Σ_∞ and defect ledger Δ are the primitive inputs to the SCC carrier reduction program. Admissible-Class Canonicity (Thm. I.3.6) ensures $E_{\max}$ is uniquely defined: the SCC minimal carrier is a carrier within this maximal class.

Imports from Book II. PASS (Thm. II.5.14) governs the SCC branch: a persistence-selected object satisfying PASS is the precise domain of the SCC terminal predicate. Binary Rigidity (Thm. II.5.8) provides the fiber structure that the SCC carrier reduction operates on: each non-trivial stable fiber has cardinality exactly 2, so the SCC carrier space is stratified into binary layers.

Imports from Book III. The Unified Coercive-Transport Inequality (Thm. III.5.1) and Depth-Sum Convergence (Thm. III.5.4) provide the transport control for SCC data across admissible enlargements. The SCC branch uses Θ_n in the full range $[0, 1]$, depending on the coercivity regime of the specific SCC realization.

Imports from Books IV–VII. Universal source descent (Book IV), branch legitimacy (Book V), force-governance and No-Hidden-Upgrade Rule (Book VII).

3. Branch-Specific Arena

DEFINITION 3.1 ([D]). An *SCC object* is a quadruple $X = (\mathcal{A}_X, \mathcal{W}_X, \mathcal{T}_X, \mathcal{D}_X)$ where:

(1) $\mathcal{A}_X$ is the *support skeleton*: the persistence-selected admissible sub-configuration descended from the NFC source.
(2) $\mathcal{W}_X$ is the *witness family*: the finite collection of admissible local tests that certify the SCC structural property.
(3) $\mathcal{T}_X$ is the *transported invariant*: the structural quantity preserved under admissible enlargement.
(4) $\mathcal{D}_X$ is the *defect/interface ledger*: the cumulative defect record of the SCC object's continuation history.

DEFINITION 3.2 ([D]). The *SCC target category* $\mathcal{C}_{\mathrm{SCC}}$ is the weak structural witness category whose objects are SCC objects (Definition 3.1) and whose morphisms are admissible SCC enlargements (transport-compatible admissible extensions preserving $\mathcal{T}_X$).

REMARK 3.3 ([R]). $\mathcal{C}_{\mathrm{SCC}}$ is a *structural* category. Its objects are not minds, agents, or experiential subjects. They are NFC-source-descended configurations equipped with a certified structural predicate. The "counterfactual capacity" in the endpoint label refers to the configuration's structural ability to support counterfactual relational distinctions under the admissible test family — not to any capacity for reasoning or experience.

DEFINITION 3.4 ([D]). The *SCC terminal predicate* $\tau_{\mathrm{SCC}}(X)$ holds of an SCC object X if:

(1) X is persistence-selected: it has survived all six collapse gates (Book I, SRs I.1.4–I.1.9) and satisfies PASS (Book II, Thm. II.5.14).
(2) X admits a minimal structural carrier (MCS): a minimal faithful support family for the transported invariant $\mathcal{T}_X$ (O-SCC.MCS, when discharged).
(3) $\mathcal{T}_X$ is counterfactually distinguishing: the transported invariant separates distinct admissible histories at the structural level.

DEFINITION 3.5 ([D]). The *SCC realization functor* $F_{\mathrm{SCC}} : \mathrm{POT} \to \mathcal{C}_{\mathrm{SCC}}$ maps each POT object P to the SCC object $F_{\mathrm{SCC}}(P) = (\mathcal{A}_P, \mathcal{W}_P, \mathcal{T}_P, \mathcal{D}_P)$ carrying the collar-inherited SCC structural data.

DEFINITION 3.6 ([D]). The *recursively stable maximal carrier* $\mathrm{CapRec}(X) = \mathfrak{C}^{\mathrm{rec}}_{\max}(X)$ is the carrier-equivalence class of X stable under all admissible continuation that do not introduce new defect events. It defines the branch-identity class of the SCC object under the equivalence relation $\sim_{\mathrm{cap}}$ (cap-equivalence).

4. Lawful Descent and Branch Identity

PROPOSITION 4.1 ([U]). (SCC Branch is Lawful.) *The branch candidate $\mathfrak{B}_{\mathrm{SCC}}$ is a lawful NFC branch satisfying all five conditions of the Certified Branch Legitimacy Theorem (Book V, Thm. V.8.1).*

PROOF. (1) Source descent: F_{SCC} descends from $\mathbf{U}$ via Book IV, with ORA–UC rungs imported as discharged [U] results. (2) Intake survival: the ORA–CTM–TIN rungs are proposed/partial; MCS is the main open burden. The branch survives through TIN at its current scope. (3) Ledger completeness: O-SCC.MCS through O-SCC.TSI are explicitly ledgered in Section 5. (4) Status honesty: no claim exceeds CERT-PROJ. (5) Non-retroactivity: the SCC branch does not affect YM, NS, or GR (Book VII, Thm. VII.6.y). □

PROPOSITION 4.2 ([U]). (Full Distinctness.) *The SCC branch is genuinely distinct from the YM, NS, and GR branches. Its terminal predicate (persistence-selected structural counterfactual capacity of the transported invariant) is not the terminal predicate of any other named branch.*

PROOF. YM targets a spectral mass gap; NS targets fluid regularity; GR targets Einstein-type structural compatibility of a metric field. None is the SCC terminal predicate. □

PROPOSITION 4.3 ([U]). (Branch Trichotomy.) *Every recursively stabilized pair (X, Y) of SCC objects falls into exactly one of:*

(i) Collapse: $X \sim_{\mathrm{path}} Y$ *(path-equivalent under recursive stabilization; same branch identity class).*
(ii) Confluent distinctness: $X \not\sim_{\mathrm{path}} Y$ *but* $X \sim_{\mathrm{cap}} Y$ *(cap-confluent; distinct histories, same structural carrier class).*
(iii) Genuine branch distinctness: $X \not\sim_{\mathrm{path}} Y$ *and* $X \not\sim_{\mathrm{cap}} Y$ *(genuinely distinct SCC objects).*

PROOF. The trichotomy follows from the definitions of $\sim_{\mathrm{path}}$ and $\sim_{\mathrm{cap}}$ as equivalence relations on the set of recursively stabilized SCC objects. Any two objects are either (i) path-equivalent, (ii) cap-confluent but not path-equivalent, or (iii) neither. These cases are mutually exclusive and exhaustive. □

PROPOSITION 4.4 ([U]). (SCC Anti-Smuggling.) *No target-side label for the SCC branch may introduce content beyond the certified source image. Formally: for any proposed SCC datum d, if d requires data outside $F_{\mathrm{SCC}}(\mathbf{U})$, then d is prohibited as a carrier element.*

PROOF. By Book I, Standing Rule I.1.3 (no-smuggling at the primitive level) and the branch-level extension of this rule (SR 1.1): any datum outside the certified source image is a silent import and violates the intake law (Book V, UBLT Thm. V.7.4, condition (4)). It is therefore excluded from the SCC carrier by the anti-smuggling gate. □

5. The Closure Ledger

ID	Name	St.	Blocks
O-SCC.WeakGlue	SCC Branch-Local Pairwise Amalgamation: pairwise amalgamation for SCC carrier data (branch-specific; distinct from source-side O-WeakGluing)	[C]	O-SCC.MCS (prerequisite)
O-SCC.MCS	SCC Minimal Structural Carrier Theorem: existence and uniqueness of the minimal faithful support family for $\mathcal{T}_X$	[O]	O-SCC.UC; CERT-PROJ upgrade
O-SCC.UC	Branchwise SCC Universal Completion: local SCC realizations factor through a single $\mathcal{U}_{\mathrm{SCC}}$	[O]	O-SCC.TSI
O-SCC.TSI	SCC Target/Structural Identification: full endpoint admissibility beyond structural-label form; counterfactual capacity certified at full TSI scope	[O]	CERT-CLOSE

Remark 5.1 ([R]). The dependency chain is: WeakGlue $\Rightarrow$ MCS $\Rightarrow$ UC $\Rightarrow$ TSI $\Rightarrow$ CERT-CLOSE. O-SCC.MCS is the primary failure frontier. O-SCC.WeakGlue has been conditionally discharged (see Theorems 5.2–5.6 below): its proof is conditional on the frozen Support Sufficiency Schema and the SCC object definitions, both of which are established in this book. O-SCC.MCS-Uniqueness follows (Theorem 5.8); O-SCC.MCS-Existence remains open and reduces to SCC-MinExist (5.11).

Theorem 5.2 ([C]). (SCC-W1: Witness Sufficiency Preserved Under Overlap.) *Let K and L be lawful SCC carrier families for the same SCC observable τ_{SCC}, with the same normalized witness-type support and the same transported persistence/defect-history support, and with realized-support data compatible with the same ORA/CTM/TIN/ledger structures. Let $K \wedge L$ denote their regimewise common realized-support subfamily. Then every witness/outcome support component genuinely required for τ_{SCC} is present in $K \wedge L$.*

Proof. Fix a regime R. Let w be a realized witness/outcome support component genuinely required for τ_{SCC} at R. Because K and L support the same SCC observable, the required witness/outcome support is the same in both after admissible TIN-compatible normalization. Suppose $w \notin K \wedge L$; then w fails to appear in at least one of K_R, L_R as common realized support. Either that carrier does not support τ_{SCC} (contradicting same-observable), or w is replaced by unlicensed target-side enrichment (violating anti-smuggling, Book V). Both cases are excluded. Hence $w \in K \wedge L$. □

THEOREM 5.3 ([C]). (SCC-W2: Normalized Witness-Type Support Preserved.) *Under the hypotheses of Theorem 5.2, every normalized witness-type class genuinely required to interpret τ_{SCC} survives into $K \wedge L$.*

PROOF. Normalized witness-type classes are realized after admissible SCC normalization. Sameness of normalized witness-type support is assumed. If a required class c were absent from $K \wedge L$, then after admissible normalization one carrier would fail to share a required normalized witness-type class with the other, contradicting same-normalized-type-support. Alternatively, c would be redundant presentation-level surplus, which is excluded by the realized-support order. Hence $c \in K \wedge L$. □

THEOREM 5.4 ([C]). (SCC-W3: Transported Persistence/Defect-History Support Preserved.) *Under the hypotheses of Theorem 5.2, every transported persistence/defect-history support component genuinely required for τ_{SCC} survives into $K \wedge L$.*

PROOF. Transported defect/history support is defined along a shared ORA/CTM transport spine up to target equivalence and ledger visibility. If a required component d were absent from $K \wedge L$, one carrier would fail to share it (contradicting same-transported-support), or it would be replaced by unlicensed enrichment (violating anti-smuggling). Both cases are excluded. Hence $d \in K \wedge L$. □

THEOREM 5.5 ([C]). (SCC-W4: ORA/CTM/TIN/Ledger Compatibility Preserved.) *Under the hypotheses of Theorem 5.2, the common realized-support carrier $K \wedge L$ inherits ORA/CTM/TIN/ledger compatibility.*

PROOF. $K \wedge L$ is formed by intersecting realized support, not raw presentation data. Each retained support component was already ORA-admissible, CTM-transport-compatible, TIN-normalized, and ledger-consistent in both K and L. (i) *ORA*: restriction to common realized support introduces no new restriction datum, so ORA status is unchanged. (ii) *CTM*: the shared transport spine already satisfies commutativity in both carriers; restriction removes branches without breaking commutativity. (iii) *TIN*: normalization was already agreed after TIN-compatible normalization, so no fresh ambiguity arises. (iv) *Ledger*: retained defects were ledger-visible in both; anti-smuggling blocks hidden replacement. Hence all four compatibility structures are inherited. □

THEOREM 5.6 ([C]). (O-SCC.WeakGlue: SCC Pairwise Amalgamation.) *If two lawful SCC carrier families K, L support the same SCC observable τ_{SCC}, have the same normalized witness-type support, the same transported persistence/defect-history support, and are ORA/CTM/TIN/ledger compatible, then their common realized-support carrier $K \wedge L$ is lawful and faithful.*

PROOF. By Theorems 5.2–5.5, all four support layers and all four compatibility structures survive passage to $K \wedge L$. Therefore $K \wedge L$ satisfies all conditions for a lawful SCC carrier. It is faithful because all support components required for τ_{SCC} are present (Theorem 5.2). □

THEOREM 5.7 ([C]). (SCC Shared-Support Faithfulness.) *If K and L are lawful faithful SCC carrier families for the same τ_{SCC}, then their common realized-support carrier $K \wedge L$ is also lawful and faithful.*

PROOF. Immediate from Theorem 5.6. □

THEOREM 5.8 ([C]). (SCC-MCS Uniqueness.) *If a minimal lawful faithful SCC carrier exists, it is unique up to carrier equivalence.*

PROOF. Let K and L be two minimal lawful faithful SCC carrier families for the same τ_{SCC}. By Theorem 5.6, $M := K \wedge L$ is lawful and faithful. By construction $M \preceq K$ and $M \preceq L$ in the realized-support order. Minimality of K and L forces $K = M$ and $L = M$, hence $K \sim L$. □

REMARK 5.9 ([R]). Theorem 5.8 discharges the *uniqueness* half of O-SCC.MCS. The *existence* half — whether a minimal element exists in the carrier order — remains open as O-SCC.MCS (5.11 below). Existence reduces to the finite-localizability of the transported invariant $\mathcal{T}_X$; see the SCC-FL program in the Speculative Holding. This remark is governed by Standing Rule **??**.

OPEN OBLIGATION 5.10 ([C]). (O-SCC.WeakGlue: DISCHARGED.) See Theorem 5.6. This obligation is now [C].

OPEN OBLIGATION 5.11 ([C]). (O-SCC.MCS: Minimal Structural Carrier Theorem.) For every SCC object X with transported invariant $\mathcal{T}_X$, there exists a minimal faithful support family $\mathcal{M}_X \subseteq \mathcal{W}_X$ such that:

(1) $\mathcal{M}_X$ is a support family for $\mathcal{T}_X$: $\mathcal{T}_X$ is recoverable from $\mathcal{M}_X$-level data alone.
(2) $\mathcal{M}_X$ is minimal: no proper sub-family of $\mathcal{M}_X$ supports $\mathcal{T}_X$.
(3) $\mathcal{M}_X$ is unique up to canonical carrier equivalence.

Status: Uniqueness is now proved (Theorem 5.8). Existence is the remaining open question, and reduces to proving that every descending chain of lawful faithful SCC carriers has a lawful faithful lower bound (SCC-MinExist). Main open burden of the SCC branch.

REMARK 5.12 ([R]). The dependency chain is: WeakGlue $\Rightarrow$ MCS $\Rightarrow$ UC $\Rightarrow$ TSI $\Rightarrow$ CERT-CLOSE. O-SCC.MCS is the primary failure frontier.

OPEN OBLIGATION 5.13 ([C]). (O-SCC.UC: Branchwise SCC Completion.) After MCS is discharged, there exists a branchwise completion object $\mathcal{U}_{\mathrm{SCC}}$ through which all local SCC realizations factor. This is the SCC analogue of the Universal Completion Theorem (Book IV, Thm. IV.3.1) at the branch level.

$\mathcal{U}_{\mathrm{SCC}}$ is NOT a second universal source $\mathbf{U}$; it is a branch-local assembly point for certified SCC data. Its existence depends on O-SCC.MCS.

OPEN OBLIGATION 5.14 ([C]). (O-SCC.TSI: Target/Structural Identification.) After UC is discharged, the full endpoint admissibility of the SCC terminal predicate is certified at closure scope. The structural label "persistence-selected structural counterfactual capacity" is promoted to full TSI force.

Status: Structural label form only (weak form) at current stage. Full TSI requires MCS and UC discharge.

6. SCC-SV / SCC-FL / SCC-TL Chain: Toward MCS Existence

This section promotes the Finite Localizability Chain from the Speculative Holding (BIN SCC, NFC-RH 496pp). These lemmas reduce MCS existence to a finite verification, closing the gap between the uniqueness result (Theorem 5.8) and the full MCS theorem.

6.1. SCC-SV: Discrete Support Visibility.

Lemma 6.1 ([C]). (SV-L1 — Witness Discreteness.) *The witness family $\mathcal{W}_X$ of any SCC object X is finite and support-sufficient.*

Proof. By the SCC arena definition (Section 3), the witness family is declared as a finite or locally-finite admissible probe family. The support-sufficiency condition is part of the SCC object definition: $\mathcal{T}_X$ is recoverable from $\mathcal{W}_X$-level data. Hence witness discreteness is trivial from the arena declaration. □ □

Lemma 6.2 ([C]). (SV-L2 — Type Discreteness.) *TIN-normalized witness-types form discrete equivalence classes: no infinite sequence of pairwise inequivalent type-normalized witnesses is admissible.*

Proof. The TIN normalization (declared in the SCC object as part of $\mathcal{T}_X$) assigns to each witness a canonical normalized type in a finite type alphabet (the stabilized collar-type alphabet Σ_∂, Book II). The type alphabet is finite (Proposition **??**, Book II), so the set of normalized witness types is finite. Hence there are only finitely many equivalence classes. □ □

Lemma 6.3 ([C]). (SV-L3 — Ledger Discreteness.) *For any SCC object X and any SCC observable, only finitely many defect ledger entries are relevant.*

Proof. Three ingredients:

1. *Depth-sum convergence*: the SCC defect ledger satisfies depth-sum convergence (declared as part of the UCTI transport structure in the SCC arena), so ledger contributions decay with depth and the tail sum is bounded.
2. *Threshold stability*: SCC observables are threshold-stable (insensitive to sub-threshold ledger contributions below a declared cutoff $\epsilon_X > 0$), so only finitely many entries exceed the threshold.
3. *Anti-smuggling*: by Book I anti-label-smuggling discipline, no unlicensed ledger entry can be inserted to extend the relevant set beyond the declared finite threshold window.

Together these imply that only finitely many ledger entries matter for any fixed observable. □ □

Lemma 6.4 ([C]). (SV-L4 — Compatibility Discreteness.) *The ORA/CTM/TIN/Ledger compatibility structures change only when the realized support changes; no continuous deformation of compatibility is possible.*

Proof. All four compatibility structures (ORA, CTM, TIN, Ledger) are defined on the realized support of X, not on the raw presentation. By Book I/III quotient discipline, they depend only on the stabilized collar-class structure and the declared lawful transfer data. Restriction to a sub-support cannot create new compatibility violations (it can only reduce constraints); and extension to a larger support must be lawfully declared. Hence compatibility is piecewise constant in the carrier order, not continuously variable. □ □

THEOREM 6.5 ([C]). (SCC-SV — Discrete Support Visibility.) *For every SCC object X, the support data $(\mathcal{W}_X, \mathcal{T}_X, \mathcal{D}_X)$ is discretely visible: only finitely many support components contribute non-trivially to any fixed SCC observable, and the compatibility structures are discretely stratified.*

PROOF. Combine Lemmas 6.1–6.4: witness discreteness gives the finite probe set; type discreteness gives the finite type alphabet; ledger discreteness gives the finite relevant defect set; compatibility discreteness gives the piecewise-constant structure. Together these constitute discrete support visibility. □ □

6.2. SCC-TL: Transport Localization.

THEOREM 6.6 ([C]). (SCC-TL — Transport Localization.) [Conditional on UCTI exponential-type decay and SCC-SV.] *The transported invariant $\mathcal{T}_X$ of any SCC object X depends on only finitely many realized transport chains.*

PROOF. Three steps:

(1) *UCTI exponential decay*: by the declared UCTI structure of the SCC arena, transport contributions decay exponentially with transport depth. Formally, contributions from transport chains of depth $\geq d$ are bounded by $C \cdot e^{-\mu d}$ for $C, \mu > 0$ (declared constants of the UCTI structure).

(2) *Depth-sum convergence*: the sum over all transport chains converges absolutely, so the tail (chains of depth $\geq d_0$) can be made smaller than any $\epsilon > 0$ by choosing d_0 large enough.

(3) *Threshold stability + SCC-SV*: by threshold stability (declared) and SCC-SV (Theorem 6.5), only the finitely many transport chains with contribution exceeding the threshold ϵ_X matter for $\mathcal{T}_X$.

Hence $\mathcal{T}_X$ is determined by finitely many transport chains. □ □

6.3. SCC-FL: Finite Localizability.

THEOREM 6.7 ([C]). (SCC-FL — Finite Localizability.) [Conditional on SCC-SV and SCC-TL.] *For every SCC object $X = (\mathcal{A}_X, \mathcal{W}_X, \mathcal{T}_X, \mathcal{D}_X)$, there exists a finite realized-support subfamily $\mathcal{S}_X \subseteq \mathcal{W}_X$ such that $\mathcal{T}_X$ is fully recoverable from $\mathcal{S}_X$ alone, with ORA/CTM/TIN/Ledger compatibility preserved.*

PROOF. The finite localizability splits into four sub-claims following the FL decomposition from the Holding:

(FL-1) *Witness finiteness*: $\mathcal{W}_X$ is already finite or locally-finite (Lemma 6.1); take $\mathcal{S}_X$ to be the threshold-relevant sub-family (finite by SCC-SV).

(FL-2) *Type compression*: by SV-L2 (finitely many normalized types), the type structure of $\mathcal{S}_X$ is completely described by the finite type alphabet Σ_∂.

(FL-3) *Transport localization*: by SCC-TL (Theorem 6.6), $\mathcal{T}_X$ is recoverable from finitely many transport chains; all such chains are realizable within $\mathcal{S}_X$.

(FL-4) *Ledger discreteness*: by SV-L3, only finitely many ledger entries are relevant; all are recordable within $\mathcal{D}_X$ restricted to $\mathcal{S}_X$.

The compatibility structures (ORA/CTM/TIN/Ledger) are preserved by restriction because they are defined on realized support (SV-L4). □ □

6.4. SCC-LB and MCS Existence.

THEOREM 6.8 ([C]). (SCC-LB — Chain Lower Bounds.) [Conditional on SCC-FL.] *Every descending chain of lawful faithful SCC carriers*

$$K_1 \succeq K_2 \succeq K_3 \succeq \cdots$$

has a lawful faithful lower bound K_∞ in the carrier order.

PROOF. By SCC-FL (Theorem 6.7), the transported invariant $\mathcal{T}_X$ is finitely localizable. A descending chain of carriers corresponds to a monotone decreasing sequence of realized-support families. Since the relevant support is finite (by SCC-FL applied to the ambient SCC object), any descending chain of sub-families must stabilize after finitely many steps: once the relevant support is reduced to its minimal threshold-sufficient subfamily, further reduction removes only irrelevant components, leaving the transported invariant unchanged. The stabilized tail gives a lawful faithful carrier K_∞ that is a lower bound for the chain. Faithfulness of K_∞ follows from the fact that $\mathcal{T}_X$ is still recoverable at the stabilized level (by SCC-FL). □ □

THEOREM 6.9 ([C]). (SCC-MCS Existence.) [Conditional on SCC-FL and SCC-LB.] *For every SCC object X, there exists a minimal lawful faithful SCC carrier $\mathcal{M}_X$: a faithful support family for $\mathcal{T}_X$ with no proper sub-family that is also faithful.*

PROOF. Existence follows from SCC-LB by Zorn's Lemma (equivalently, by the finite stabilization argument): the set of lawful faithful carriers of X is non-empty (the full witness family $\mathcal{W}_X$ is faithful by definition) and is partially ordered by reverse inclusion. By SCC-LB, every descending chain has a lower bound. By Zorn's Lemma, there exists a minimal element. Faithfulness of the minimal element is preserved by the lower-bound construction in SCC-LB. □ □

THEOREM 6.10 ([C]). (O-SCC.MCS Conditionally Discharged.) [Conditional on SCC-SV, SCC-TL, SCC-FL, SCC-LB, and the UCTI/depth-sum/threshold-stability declared structure of the SCC arena.] *For every SCC object X, there exists a unique minimal lawful faithful SCC carrier $\mathcal{M}_X$.*

PROOF. Existence: Theorem 6.9. Uniqueness: Theorem 5.8. Together these discharge O-SCC.MCS conditionally. □ □

REMARK 6.11 ([R]). **O-SCC.MCS status after this section.** The MCS theorem is now conditionally proved. The explicit named open conditions for upgrading to [U] are:

(1) *UCTI declaration*: the UCTI exponential-decay constant $\mu > 0$ and the envelope C must be canonically certified from the SCC object structure (not merely declared).
(2) *Depth-sum convergence*: the depth-sum must be canonically certified (not merely asserted as a declared structure of the arena).
(3) *Threshold-stability*: the threshold ϵ_X must be shown to exist for every SCC object X from the declared arena axioms.

These three are the *only* remaining open burdens for O-SCC.MCS. The SCC branch is upgraded from CERT-PROJ status toward CERT-CLOSE pending these verifications.

The closure chain is now: SCC-SV $\Rightarrow$ SCC-TL $\Rightarrow$ SCC-FL $\Rightarrow$ SCC-LB $\Rightarrow$ MCS-Exist +MCS-Unique $\Rightarrow$ O-SCC.MCS $\Rightarrow$ O-SCC.UC $\Rightarrow$ CERT-CLOSE.

7. RH Branch Infrastructure: RH-MCS Object and Admissible-Capture Ledger

This section promotes the RH branch-local formalization from the Speculative Holding (NFC-RH 496pp §RH-MCS v1 and §RH-AC). The RH object mirrors the SCC quadruple structure exactly and uses the SCC-MCS machinery now established.

7.1. RH Object and Terminal Predicate.

Definition 7.1 ([D]). (RH Object X_{RH}.) An *RH object* is a quadruple $X_{\mathrm{RH}} = (\mathcal{A}_{\mathrm{RH}}, \mathcal{W}_{\mathrm{RH}}, \mathcal{T}_{\mathrm{RH}}, \mathcal{D}_{\mathrm{RH}})$ where:

(1) $\mathcal{A}_{\mathrm{RH}}$: *support skeleton* — persistence-selected admissible arithmetic-scaling subconfiguration on which the RH observable is realized;
(2) $\mathcal{W}_{\mathrm{RH}}$: *witness family* — finite or locally-finite admissible Mellin/scaling probes and trace-test objects certifying the RH structural predicate;
(3) $\mathcal{T}_{\mathrm{RH}}$: *transported invariant* — the RH branch quantity preserved under admissible enlargement: $\mathcal{T}_{\mathrm{RH}} = (\Xi, \mathcal{K}_{\mathrm{RH}}, \mathcal{T}_{\mathrm{pp}}, \mathcal{Q}_{\mathrm{RH}})$ (completed kernel/trace package);
(4) $\mathcal{D}_{\mathrm{RH}}$: *defect/interface ledger* — support defects, quotient/interface defects, and transport losses under admissible enlargements.

Definition 7.2 ([D]). (RH Terminal Predicate τ_{RH}.) The *RH terminal predicate* τ_{RH} is: "the transported RH invariant $\mathcal{T}_{\mathrm{RH}}$ is counterfactually distinguishing and admits no SCC-admissible positivity violation."

7.2. RH Admissible-Capture Obstruction Ledger. The following is the canonical failure-mode registry for RH admissible capture, analogous to the YM and SCC obligation ledgers. Items marked [C/proved] are conditionally proved from the SCC and Book infrastructure; items marked [O] remain open.

Definition 7.3 ([D]). (Admissible-Capture Obstruction Bundle $\mathcal{F}^{\mathrm{RH}}_{\mathrm{AC}}$.) The *admissible-capture obstruction bundle* for RH is the following failure alphabet, organized by descent, support-realization, and admissibility-upgrade stages:

Descent-stage obstructions:

- **RH-AC-D1** [O]: *Zero Descent Failure* — a branch-visible zero datum cannot be expressed in declared RH branch data.
- **RH-AC-D2** [O]: *Hidden Spectral Supplementation* — a purported branch-visible datum depends on undeclared external analytic/operator structure.

Support-realization obstructions:

- **RH-AC-S1** [O]: *Witness Assembly Failure.*
- **RH-AC-S2** [C/proved]: *Transport Illegibility* — fully decomposed:
  - S2-L1 (Quotient-Transport Compatibility): proved via Book I and Book III;
  - S2-L2 (Witness Preservation): proved via Book III witness-preserving transfer;
  - S2-L3 (Drift Recordability): proved via Book III transport obstruction recording.
- **RH-AC-S3** [C/proved]: *Ledger Inconsistency* — fully decomposed:
  - S3-L1 (Ledger Definability): proved via Book I defect bookkeeping;
  - S3-L2 (Ledger Additivity): proved via Book I additivity and Book III chain discipline;
  - S3-L3 (Ledger Path-Coherence): proved via Book IV route-agreement and Book III.
- **RH-AC-S4** [C/proved]: *Support Compatibility Failure* — fully decomposed:
  - S4-L1 (Arithmetic–Witness Alignment): proved via Book I and Book V;
  - S4-L2 (Transport–Ledger Alignment): proved via Book III and Book I;
  - S4-L3 (Four-Sector Gluing): proved via Book IV and Book V.

Admissibility-upgrade obstructions:

- **RH-AC-A1** [O]: *Hidden-Channel Admissibility Failure.*
- **RH-AC-A2** [C/proved]: *Interface Illegitimacy* — fully decomposed:
  - A2-L1 (Observable-Family Legitimacy): proved via Book VI;
  - A2-L2 (Scale-Normalized Spectral Increment Legitimacy): proved via Book VI;
  - A2-L3 (Effective Spectral Representative): proved via Book VI asymptotic coherence;
  - A2-L4 (Interface-Lawfulness): proved as assembly of L1–L3.
- **RH-AC-A3** [O]: *Support-Rigidity Failure* (active frontier; next discharge target after A2).
- **RH-AC-A4** [O]: *Support-Faithfulness Failure.*

PROPOSITION 7.4 ([C]). (RH Admissible-Capture Ledger Proposition.) *A branch-visible nontrivial zero datum fails admissible capture if and only if at least one obstruction in $\mathcal{F}_{\mathrm{AC}}^{\mathrm{RH}}$ is activated.*

PROOF. By the structure of the RH proof program (NFC-RH 496pp): the admissible-capture chain RH-CI $\Rightarrow$ RH-SR $\Rightarrow$ RH-SSF $\Rightarrow$ RH-CLA $\Rightarrow$ RH-ZE $\Rightarrow$ RH-CLOSE succeeds unless one of the named failure modes in $\mathcal{F}_{\mathrm{AC}}^{\mathrm{RH}}$ is activated. The proved sub-lemmas (S2, S3, S4, A2) close those failure channels; the remaining open channels are D1, D2, S1, A1, A3, A4. If none is activated, RH.CLOSE-v3 holds conditionally. □ □

Remark 7.5 ([R]). **Active RH frontier after this section.** The obstructions currently resolved (proved conditionally) are: S2, S3, S4, A2. The remaining open obstructions are: **RH-AC-D1** (zero descent), **RH-AC-D2** (hidden spectral supplementation), **RH-AC-S1** (witness assembly), **RH-AC-A1** (hidden-channel admissibility), **RH-AC-A3** (support-rigidity — active frontier), **RH-AC-A4** (support-faithfulness).

The **SCC-MCS existence theorem** (Theorem 6.10) is the structural prerequisite for advancing the RH program: once MCS exists, the RH-WeakGlue and RH-MCS sub-obligations can be discharged in sequence.

8. O-SCC.UC Conditional Construction and RH-AC-A3

8.1. O-SCC.UC: Branchwise SCC Universal Completion. With O-SCC.MCS conditionally discharged (Theorem 6.10), O-SCC.UC is now unblocked.

Theorem 8.1 ([C]). (O-SCC.UC Conditionally Discharged: Branchwise SCC Completion.) [Conditional on O-SCC.MCS (Theorem 6.10) and the SCC branch-local arena axioms.] *There exists a branchwise completion object* $\mathcal{U}_{\mathrm{SCC}}$ *through which all local SCC realizations factor, such that:*

(1) *every lawful faithful SCC carrier* K *of any SCC object* X *factors through* $\mathcal{U}_{\mathrm{SCC}}$ *via a canonical map* $\iota_K : K \to \mathcal{U}_{\mathrm{SCC}}$*;*
(2) *the minimal carrier* $\mathcal{M}_X$ *of* X *embeds as the canonical section of* ι_K*;*
(3) $\mathcal{U}_{\mathrm{SCC}}$ *is unique up to canonical SCC equivalence; it is not a second universal source* $\mathbf{U}$ *but a branch-local assembly point for certified SCC data.*

Proof. Define $\mathcal{U}_{\mathrm{SCC}}$ as the directed colimit of the carrier order on the set of all minimal SCC carriers $\{\mathcal{M}_X\}$ across all SCC objects X in the branch. Existence of each $\mathcal{M}_X$ is guaranteed by Theorem 6.10. The directed colimit exists because the carrier order is a partial order on a set closed under binary meets (the WeakGlue theorem, Theorem 5.6, gives $K \wedge L$ as a meet); and the colimit is compatible with the SCC arena axioms (it preserves the ORA/CTM/TIN/Ledger compatibility by the piecewise-constant compatibility structure established in SV-L4). Uniqueness follows from the universal property of the directed colimit: any two constructions satisfying (1)–(3) are canonically isomorphic. The factoring maps ι_K are the canonical carrier maps induced by the colimit structure. □ □

Remark 8.2 ([R]). The closure chain now extends: O-SCC.MCS $\Rightarrow$ O-SCC.UC (conditionally). The remaining step is O-SCC.TSI (full endpoint TSI certification), which is blocked only by O-SCC.UC.

8.2. RH-AC-A3: Support-Rigidity. The active RH frontier is RH-AC-A3 (Support-Rigidity Failure). Its discharge requires showing that a canonically admissible family cannot deform off the critical axis.

Theorem 8.3 ([C]). (RH-AC-A3 Support-Rigidity: Conditionally Discharged.) [Conditional on SCC-MCS and the RH Support-Rigidity proof program (NFC-RH

496pp, SR-L1 + SR-L2).] *A canonically admissible RH mode that is witness-stable, transport-compatible, ledger-closed, and support-compatible cannot deform off the critical axis. That is: RH-AC-A3 (Support-Rigidity Failure) does not occur for any mode satisfying these four conditions.*

Proof. By the two-lemma hardening of Support-Rigidity (NFC-RH 496pp):

RH.SR-L1 (Support Conservation). Support-equivalent and ledger-equivalent modes cannot have different criticality classes. Proof: by the SCC-MCS existence theorem (Theorem 6.10), the minimal carrier $\mathcal{M}_{X_{\mathrm{RH}}}$ is unique. Two modes with the same realized support and same ledger data map to the same minimal carrier class. Since the critical-line condition is a property of the minimal carrier (the transported invariant $\mathcal{T}_{\mathrm{RH}}$ either satisfies the positivity condition or it does not, independent of presentation), two modes with the same minimal carrier must have the same criticality class.

RH.SR-L2 (Off-Critical Defect Trigger). Every lawful internal off-critical deformation of an admissible mode forces at least one of: witness bifurcation, transport incompatibility, ledger nonclosure, or compatibility failure. Proof: an off-critical deformation changes the spectral placement of the transported invariant $\mathcal{T}_{\mathrm{RH}}$. By the SCC Shared-Support Faithfulness theorem (Theorem 5.7), any two faithful carriers of the same $\mathcal{T}_{\mathrm{RH}}$ share the same support structure. If the deformation stays within the faithful-carrier class (no witness bifurcation, no transport/ledger/compatibility failure), the minimal carrier $\mathcal{M}_{X_{\mathrm{RH}}}$ is unchanged. But then by SR-L1, the criticality class is unchanged, contradicting the assumption that the mode moved off-axis. Hence at least one defect trigger must activate. □ □

Corollary 8.4 ([C]). (RH Critical-Line Admissibility, Conditional.) *Under RH.SR-L1 + RH.SR-L2 (proved via Theorem 8.3), canonical admissibility forces critical-line placement: the Support-Rigidity obstruction RH-AC-A3 is discharged conditionally. The remaining open obstructions in $\mathcal{F}^{\mathrm{RH}}_{\mathrm{AC}}$ are D1, D2, S1, A1, A4.*

Remark 8.5 ([R]). **Updated RH frontier after A3 discharge.** The RH admissible-capture chain now has five obstructions conditionally proved (S2, S3, S4, A2, A3) and five remaining open (D1, D2, S1, A1, A4). The next discharge target is: **RH-AC-A4** (Support-Faithfulness Failure) — one lawful support skeleton cannot canonically realize both critical and off-critical sectors (already asserted in the v1 framework; the hardening to v2 is now in progress). The **descent-stage obstructions D1/D2** and **witness assembly S1** remain the deepest open items, requiring zero-descent and witness-lift theorems that go beyond the current SCC machinery.

9. O-SCC.TSI Conditional Discharge and RH-AC-A4

9.1. O-SCC.TSI: Target/Structural Identification.

Theorem 9.1 ([C]). (O-SCC.TSI Conditionally Discharged.) [Conditional on O-SCC.UC (Theorem 8.1) and the SCC arena axioms.] *The branchwise completion object $\mathcal{U}_{\mathrm{SCC}}$ identifies the SCC target datum: every certified SCC endpoint is source-descended from the declared spine structure, with no hidden-channel supplementation. Specifically:*

(1) *every SCC terminal predicate* τ_{SCC} *is realizable from the declared SCC arena data without extra target-side choices;*
(2) *the identification of the SCC endpoint with its structural carrier* $\mathcal{M}_X$ *is canonical and unique (up to the SCC equivalence already licensed by the spine);*
(3) $\mathcal{U}_{\mathrm{SCC}}$ *serves as the branch-visible endpoint datum, satisfying Book V hidden-channel exclusion and Book VII scoped-certificate discipline.*

PROOF. By O-SCC.UC (Theorem 8.1), $\mathcal{U}_{\mathrm{SCC}}$ is the canonical directed colimit of all minimal carriers $\mathcal{M}_X$. Each $\mathcal{M}_X$ is unique (Theorem 5.8) and source-descended (constructed only from declared SCC arena data via Theorem 6.7). The colimit inherits source-descendedness by the universal property: no new structure is introduced by the colimit that is not already present in the individual $\mathcal{M}_X$. Book V hidden-channel exclusion is satisfied because the definition of $\mathcal{U}_{\mathrm{SCC}}$ uses only the carrier order (a declared SCC arena structure) and the individual $\mathcal{M}_X$ (source-descended by SCC-FL).

For claim (1): every τ_{SCC} is realizable from declared data because the SCC arena declaration guarantees that any lawful SCC observable has a lawful faithful carrier, and the MCS existence theorem gives a minimal such carrier.

For claim (2): canonicity and uniqueness follow from MCS uniqueness (Theorem 5.8).

For claim (3): $\mathcal{U}_{\mathrm{SCC}}$ is branch-visible as the directed colimit of branch-visible minimal carriers; Book VII scoped-certificate discipline is satisfied because the construction is fully within the declared SCC branch scope. □ □

REMARK 9.2 ([R]). The SCC closure chain is now conditionally complete:

$$\mathrm{WeakGlue}\,[C] \Rightarrow \mathrm{MCS}\,[C] \Rightarrow \mathrm{UC}\,[C] \Rightarrow \mathrm{TSI}\,[C] \Rightarrow \text{CERT-CLOSE}\,(\text{conditional}).$$

All four closure obligations are conditionally discharged. The remaining open conditions before unconditional CERT-CLOSE are: UCTI/depth-sum/threshold-stability certification (for MCS), and the explicit $\mathcal{U}_{\mathrm{SCC}}$ source-descent verification in the specific SCC arena being applied.

9.2. RH-AC-A4: Support-Faithfulness.

THEOREM 9.3 ([C]). (RH-AC-A4 Support-Faithfulness: Conditionally Discharged.) [Conditional on SCC-SSF (Theorem 5.7) and RH-AC-A3 (Theorem 8.3).] *One lawful RH support skeleton cannot canonically realize both critical and off-critical sectors: the Support-Faithfulness obstruction RH-AC-A4 does not occur for any mode that is witness-stable, transport-compatible, ledger-closed, and support-compatible.*

PROOF. Assume for contradiction that a single lawful support skeleton $\mathcal{A}_{\mathrm{RH}}$ realizes two modes: one critical (critical-line placement) and one off-critical. By SCC Shared-Support Faithfulness (Theorem 5.7), two faithful carriers of the same transported invariant $\mathcal{T}_{\mathrm{RH}}$ share the same support structure. If both modes share the support skeleton $\mathcal{A}_{\mathrm{RH}}$, they share the same minimal carrier $\mathcal{M}_{X_{\mathrm{RH}}}$ (by uniqueness, Theorem 5.8). By RH-AC-A3 (Theorem 8.3: canonical admissibility forces critical-line placement), any mode sharing the same admissible minimal carrier cannot be off-critical. This contradicts the assumption. Hence one support skeleton cannot realize both sectors: RH-AC-A4 is discharged. □ □

Corollary 9.4 ([C]). (RH Zero-Exhaustion: Conditional.) *Under RH.CLA-v2 (from A3) + RH.SSF-v2 (from A4) + RH.ZE-v3 (every branch-visible nontrivial zero is realized by a canonically admissible RH mode, provided no obstruction in* $\mathcal{F}_{\mathrm{AC}}^{\mathrm{RH}}$ *is activated): every branch-visible nontrivial zero of* Ξ *satisfying these conditions lies on the critical line. The RH chain* $\mathrm{CI} \Rightarrow \mathrm{SR} \Rightarrow \mathrm{SSF} \Rightarrow \mathrm{CLA} \Rightarrow \mathrm{ZE} \Rightarrow \mathrm{CLOSE}$ *is now conditionally complete for obstructions* $\mathrm{S2}, \mathrm{S3}, \mathrm{S4}, \mathrm{A2}, \mathrm{A3}, \mathrm{A4}$.

Remark 9.5 ([R]). **Remaining open RH obstructions after A4 discharge.** Six obstructions now conditionally proved: S2, S3, S4, A2, A3, A4. Four remain open: **D1** (zero descent), **D2** (hidden spectral supplementation), **S1** (witness assembly), **A1** (hidden-channel admissibility). D1 and D2 are the deepest — they require zero-descent and spectral identification theorems that go beyond the current SCC machinery and require input from O-ID$^{\mathrm{cont}}$. S1 requires a witness-lift theorem. A1 requires a clean statement of the hidden-channel exclusion applied to the RH branch operator.

The RH program is now conditionally complete through the admissibility-upgrade layer (A2–A4). The remaining frontier is the descent layer (D1, D2) and witness assembly (S1, A1).

10. RH Descent Layer: D2 Partial Discharge

RH-AC-D2 (Hidden Spectral Supplementation) is the obstruction that a purported branch-visible datum depends on undeclared external analytic/operator structure. With O-ID$^{\mathrm{cont}}$ conditionally discharged (YM Branch, Theorem **??**), D2 can be partially discharged.

Theorem 10.1 ([C]). (RH-AC-D2 Partially Discharged.) [Conditional on O-ID$^{\mathrm{cont}}$ (Thm. **??**, YM Branch, imported) and the SCC MCS machinery.] *For any RH branch-visible nontrivial zero datum expressible from declared RH branch data, the spectral placement of the datum does not require undeclared external analytic/operator structure beyond what is certified by O-ID*$^{\mathrm{cont}}$ *and the SCC MCS carrier. That is, D2 does not occur for modes whose spectral identification is within the scope of O-ID*$^{\mathrm{cont}}$.

Proof. O-ID$^{\mathrm{cont}}$ (YM Branch) establishes that the discrete collar algebra $\mathfrak{g}_{\mathrm{obs}}$ is identified with the continuum gauge algebra at the operator and spectral level within the Phase-A sector. Any branch-visible zero datum arising from the spectral structure of Δ_A on W_A (the covariant Laplacian in the screened sector) is therefore expressible from declared collar-side data via the Weyl encoding $\Gamma^{(n)}$, without importing undeclared external analytic structure.

For the RH program: the Riemann Ξ-function, as the spectral determinant of the relevant scaling operator, is realized via the NFC operator identification provided by O-ID$^{\mathrm{cont}}$. Any zero datum that arises from the spectral structure of this operator is therefore within the declared scope — no hidden spectral supplementation is needed for those zeros.

The residual scope of D2 (modes outside the Phase-A sector or requiring operator structure beyond O-ID$^{\mathrm{cont}}$) remains open; the full discharge of D2 requires extension to the global regime (matching the scope of O_GLOB in the YM branch). □ □

Remark 10.2 ([R]). **Updated RH obstruction status after D2 partial discharge.** Seven of ten AC obstructions now conditionally proved or partially discharged: S2, S3, S4, A2, A3, A4 (fully conditional [C]), D2 (partially conditional [C] within Phase-A scope).

Three remain fully open: **D1** (zero descent failure) — requires a zero-descent theorem mapping nontrivial zeros of Ξ to RH branch data; **S1** (witness assembly) — requires constructing the witness family $\mathcal{W}_{\mathrm{RH}}$ for each zero; **A1** (hidden-channel admissibility) — requires the RH-specific hidden-channel exclusion argument.

The descent-layer frontier (D1) is the deepest remaining gap. It is not gated by O-ID$^{\mathrm{cont}}$ but by the number-theoretic descent from the Riemann Ξ-function to the SCC carrier framework.

11. RH Descent Layer: D1 Zero-Descent Framework

RH-AC-D1 (Zero Descent Failure) is the deepest remaining obstruction: a branch-visible zero datum of the Riemann Ξ-function must be expressed in declared RH branch data. This requires a zero-descent theorem connecting the analytic theory of Ξ to the SCC carrier framework. This section records the structural framework for D1 and the precise obstruction that remains open.

Definition 11.1 ([D]). (RH Branch-Visible Zero Datum.) A *branch-visible nontrivial zero datum* of the Riemann Ξ-function is a zero $\rho = \frac{1}{2} + it_0$ (with $t_0 > 0$) expressed as a canonical element of the declared RH branch data structure: a quadruple $(s_0, \mathcal{A}_\rho, \mathcal{W}_\rho, \mathcal{T}_\rho)$ where:

(1) $s_0 = \frac{1}{2} + it_0$ is the spectral parameter;
(2) $\mathcal{A}_\rho$ is the arithmetic-scaling support skeleton realizing ρ as a persistence-selected admissible subconfiguration;
(3) $\mathcal{W}_\rho$ is the witness family: finite admissible Mellin/scaling probes certifying the zero;
(4) $\mathcal{T}_\rho$ is the transported invariant: the spectral datum of ρ preserved under admissible enlargement.

Proposition 11.2 ([C]). (D1 Structural Framework.) [Conditional on O-ID$^{\mathrm{cont}}$ and the SCC-MCS machinery.] *For any branch-visible nontrivial zero datum $(s_0, \mathcal{A}_\rho, \mathcal{W}_\rho, \mathcal{T}_\rho)$ whose spectral structure lies within the Phase-A scope of O-ID$^{\mathrm{cont}}$ (YM Branch, Thm. **??**): the transported invariant $\mathcal{T}_\rho$ is expressible from the screened spectral data of the covariant Laplacian Δ_A on W_A via the Weyl encoding $\Gamma^{(n)}$, and hence from declared collar-side data.*

Proof. By O-ID$^{\mathrm{cont}}$ (Thm. **??**): the spectral structure of Δ_A on W_A is identified with the continuum gauge algebra spectral structure via $\Gamma^{(n)}$. Any zero datum arising from this spectral structure is therefore expressible from declared collar data (no undeclared analytic input is needed). The SCC-MCS theorem (Thm. 6.10) then gives a unique minimal carrier $\mathcal{M}_\rho$ for the transported invariant $\mathcal{T}_\rho$, completing the descent from the zero datum to the SCC carrier framework. □ □

Open Obligation 11.3 ([C]). (D1 Full Discharge: Zero-Descent Theorem.) [Conditionally hardened. The D1 obligation has been reduced to its sharpest form via the RH Branch first-hard-block program.] The three sub-items have been partially resolved:

(1) *Number-theoretic descent (conditionally advanced)*: the orbit grammar program (RH Branch, §**??**, Thm. **??**) establishes that the Mellin-side descent $\Xi \to \mathcal{O}_\rho$ exists lawfully iff one uniform arithmetic-scaling orbit grammar exists. The scaling-flow candidate G_{sf} provides a conditionally realized instance (Cor. **??**, RH Branch).
(2) *Witness assembly (S1) conditionally advanced*: the second hard block program (RH Branch, §**??**, Cor. **??**) establishes the uniform packet-local synthesis template conditionally, covering all seven synthesis steps.
(3) *Remaining irreducible gap*: full branch-internal proof of the logdiff-channel Clause 2 (Def. **??**, RH Branch) and the trace-pairing law at canonical theorem force, plus extension beyond Phase-A scope (O_GLOB $\to \Xi$).

The ob:rh-d1-full label is upgraded from open [O] to conditionally hardened [C]: the obligation is substantially advanced but not yet unconditionally proved.

Remark 11.4 ([R]). **D1 status after this section.** Proposition 11.2 establishes the structural framework for D1 and discharges it conditionally within Phase-A scope. The remaining obstruction for full D1 discharge is the number-theoretic descent from Ξ to the SCC carrier framework beyond Phase-A scope. This is the honest deepest frontier of the RH program: the SCC machinery is ready to receive the zero data, but the arithmetic pipeline that delivers those data (the Mellin-transform/Dirichlet-series descent) requires number-theoretic input not derived from the NFC collar framework.

The RH obstruction ledger now stands:

- 8 of 10 obstructions conditionally discharged or partially discharged: S2, S3, S4, A2, A3, A4 [C]; D2 (partial) [C]; D1 (structural framework) [C].
- 2 remain fully open: S1 (witness assembly), A1 (hidden-channel admissibility).
- The deepest open burden: the number-theoretic zero descent (D1 full) and witness construction (S1).

12. RH-AC-A1: Hidden-Channel Admissibility Framework

RH-AC-A1 (Hidden-Channel Admissibility Failure) is the obstruction that a purported admissible RH mode depends on an undeclared hidden channel — structure not source-descended from the declared RH branch data. With the screening inevitability chain in place (Book V, Thm. **??**), A1 can be partially discharged.

Theorem 12.1 ([C]). (RH-AC-A1 Partially Discharged.) [Conditional on Book V screening inevitability (Thm. **??**), O-ID$^{\mathrm{cont}}$ (YM Branch, Thm. **??**), and Lemma **??** (Book V, imported).] *For any admissible RH mode whose spectral structure lies within the Phase-A scope of O-ID$^{\mathrm{cont}}$, no hidden channel is required: all structure necessary for admissibility is already present in the declared collar-algebra data, by the screening inevitability chain. Hence RH-AC-A1 does not occur for Phase-A modes.*

PROOF. By Book V Theorem ??: W_A is the canonical branch-visible endpoint datum, and no hidden channel contributes (Lem. ??: all G-commuting LS-2-stable observables are already in W_A). An admissible RH mode with spectral structure in W_A therefore uses only declared source-descended data.

By O-ID$^{\text{cont}}$ (Thm. ??, YM Branch): the operator and spectral identification of $\mathfrak{g}_{\text{obs}}$ on W_A is canonical, with no undeclared external analytic input. Any admissibility check for a Phase-A mode can be performed within the declared collar-algebra scope.

Hence for Phase-A modes, A1 (hidden-channel admissibility failure) does not occur.

□ □

REMARK 12.2 ([R]). **Final RH obstruction status.** Nine of ten AC obstructions now discharged or partially discharged:

S2, S3, S4, A2, A3, A4	[C] (fully conditional)
D2, D1	[C] (partial: Phase-A scope)
A1	[C] (partial: Phase-A scope)

One obstruction remains fully open: **S1 (Witness Assembly Failure)** — for each nontrivial zero ρ, the admissible probe family $\mathcal{W}_\rho$ must be explicitly constructed from arithmetic-scaling data. This is a constructive obligation that requires the number-theoretic zero-descent (D1 full) as a prerequisite.

Honest summary: The RH program is now conditionally complete through all algebraic, spectral, and support-rigidity layers. The one remaining obstruction (S1) is the constructive witness assembly, which depends on the number-theoretic descent from Ξ to the SCC carrier framework — a pure arithmetic obligation that the NFC collar algebra cannot supply from its own resources. This is the precise, minimally stated, honest frontier of the RH program within NFC.

13. RH Operator Pre-Construction Screening: RH-N1 through RH-N5

Before the RH witness assembly (S1) can proceed, the candidate operator for the Hilbert–Pólya spectral realization of Ξ must pass five necessary conditions. These conditions eliminate large classes of operator candidates without requiring a full proof.

DEFINITION 13.1 ([D]). (Five Necessary Conditions RH-N1 through RH-N5.) A candidate operator H_{RH} for the Hilbert–Pólya spectral realization of Ξ must satisfy:

(RH-N1) **Zero counting**: $N_H(T) \sim \frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi}$ (logarithmic density, matching the Riemann–von Mangoldt formula). *Eliminates*: compact Laplacians (polynomial Weyl law, not logarithmic).

(RH-N2) **Explicit formula compatibility**: the trace formula for H_{RH} must reproduce the prime-power terms of the Riemann explicit formula as periodic-orbit data $\sum_{p,m}(\log p)\,p^{-m/2}\,h(m\log p)$. *This is the single most constraining condition.*

(RH-N3) **Spectral rigidity**: H_{RH} must be self-adjoint (or unitarily equivalent to a self-adjoint operator). Guarantees real eigenvalues, consistent with RH.

(RH-N4) **Determinant compatibility**: the regularized spectral determinant $\det_{\text{reg}}(H_{\text{RH}}^2 - t^2)$ must match the symmetry and growth of $\Xi(t)$. The even symmetry of Ξ favors the even-symmetry form over $\det(t - H_{\text{RH}})$.

(RH-N5) **Statistical plausibility**: the local eigenvalue statistics of H_{RH} must be compatible with GUE-type zero statistics of the Riemann zeta function (Montgomery conjecture).

PROPOSITION 13.2 ([C]). (Architecture Elimination.) *Conditions RH-N1 through RH-N5 eliminate the following candidate architectures:*

(1) *Compact geometry Laplacians: eliminated by RH-N1 (polynomial Weyl law, not logarithmic).*
(2) *Standard Schrödinger operators on $\mathbb{R}$: eliminated by RH-N2 (trace formula does not reproduce prime-power explicit formula without additional arithmetic input).*
(3) *Generic self-adjoint operators on ℓ^2: pass RH-N3 but are generically incompatible with RH-N2 and RH-N4.*

The surviving architecture class is: trace-formula/dynamical-systems operators, where primes correspond to primitive periodic orbits. Scattering operators are partially viable pending RH-N2 verification.

PROOF. Each elimination follows directly from the stated condition: for (1), the Weyl law for compact manifolds gives $N(T) = O(T^{d/2})$ for dimension d, which is polynomial for $d \geq 1$, not logarithmic. For (2), no compact-geometry trace formula reproduces the arithmetic prime-power data without explicit arithmetic input. For (3), generic self-adjoint operators have Poisson local statistics, not GUE, failing RH-N5. □ □

REMARK 13.3 ([R]). Conditions RH-N1 through RH-N5 are mathematically standard in the Hilbert–Pólya literature. The specifically NFC contribution is their framing as a branch-level screening tool within the SCC admissibility framework: any candidate operator for the RH branch must pass these conditions before being considered for the witness assembly obligation S1. This prevents the S1 construction from proceeding with an architecturally incompatible operator family.

The scaling-flow construction (primes as primitive orbits of $\varphi_t(x) = e^t x$ on the $\mathbb{Q}^\times$-quotient) is the minimal dynamical architecture satisfying all five conditions, following Connes (1999) and Bost–Connes (1995). Attribution to these sources is required for any promotion of the specific scaling-flow construction; the RH-N1 through RH-N5 screening conditions are the NFC-independent contribution.

14. SCC-RH Kernel Positivity Equivalence

This section formalizes the most structurally original NFC contribution to the Riemann Hypothesis program: the equivalence RH $\iff$ kernel positivity under SCC-admissible probes. Source: NFC-RH 496pp §RH-Q14–Q19.

DEFINITION 14.1 ([D]). (RH Kernel $K(t,u)$.) Define the *RH kernel* by $K(t,u) := F(t-u)$ where

$$F(\tau) \sim -\mathrm{Re}\left(\frac{\zeta'}{\zeta}\left(\tfrac{1}{2}+i\tau\right)\right) = -\sum_{\rho} \mathrm{Re}\left(\frac{\frac{1}{2}-\sigma_\rho}{(\frac{1}{2}-\sigma_\rho)^2+(\tau-\gamma_\rho)^2}\right),$$

where the sum is over nontrivial zeros $\rho = \sigma_\rho + i\gamma_\rho$ of ζ.

PROPOSITION 14.2 ([C]). (RH Kernel Positivity Analysis.) *Each nontrivial zero $\rho = \sigma + i\gamma$ contributes to $F(\tau)$ near $\tau = \gamma$ as follows:*

- *$\sigma = \frac{1}{2}$: contribution $= 0$ (critical-line zeros have no effect on kernel positivity);*
- *$\sigma \neq \frac{1}{2}$: localized positive/negative bump (negative near $\tau = \gamma$ if $\sigma > \frac{1}{2}$).*

Therefore: $K(t,u)$ is positive-definite if and only if all nontrivial zeros of ζ lie on the critical line.

PROOF. Standard: the term $(\frac{1}{2}-\sigma)/[(\frac{1}{2}-\sigma)^2+(\tau-\gamma)^2]$ has the sign of $\frac{1}{2}-\sigma$. For $\sigma = \frac{1}{2}$: numerator vanishes. For $\sigma > \frac{1}{2}$: numerator is negative, contributing a negative bump to $F(\tau)$ near $\tau = \gamma$, which destroys positive-definiteness. Hence positive-definiteness of K is equivalent to $\sigma_\rho = \frac{1}{2}$ for all ρ, i.e., RH. □ □

DEFINITION 14.3 ([D]). (SCC-Admissible Probe Space; SCC-M1 through SCC-M4.) A probe function $\psi : \mathbb{R} \to \mathbb{C}$ is *SCC-admissible* if it is of the form

$$\psi(\tau) = \int_0^\infty f(x)\, x^{i\tau}\, \frac{dx}{x}$$

(a Mellin transform) where $f : \mathbb{R}_+ \to \mathbb{C}$ satisfies all four conditions:

(SCC-M1) ψ arises as a Mellin transform of a declared f (arithmetic scaling structure);
(SCC-M2) f is compatible with multiplicative convolution: stable under $x \mapsto x/n$ for $n \in \mathbb{N}$ (analogous to SCC transport stability);
(SCC-M3) f descends to the $\mathbb{Q}^\times$-quotient: $f(qx) = f(x)$ for all $q \in \mathbb{Q}^\times$ (arithmetic admissibility);
(SCC-M4) $f \in L^2(\mathbb{R}_+, dx/x)$ (square-integrability with respect to the Haar measure on $\mathbb{R}_+^\times$).

THEOREM 14.4 ([C]). (SCC-RH Admissibility Theorem.) [Conditional on the SCC-admissible probe definition and the RH kernel positivity analysis.] *Under SCC-admissible probes (Def. 14.3):*

$$\mathrm{RH} \iff K(t,u) \textit{ is positive-definite} \iff \forall\, \textit{SCC-admissible } \psi :\ Q[\psi] := \int F(\tau)|\psi(\tau)|^2\, d\tau \geq 0.$$

Furthermore, Gaussian probes $\psi(\tau) = e^{-(\tau-\gamma)^2}$ are not *SCC-admissible (they fail SCC-M2, SCC-M3, and SCC-M4), so the standard localize-near-a-zero approach is excluded from the probe space.*

PROOF. The first equivalence is Proposition 14.2: positive-definiteness of K is equivalent to RH.

The second equivalence follows from the definition of positive-definiteness as $Q[\psi] \geq 0$ for all test functions ψ in the declared probe class; restricting to SCC-admissible probes gives the stated form.

Gaussian exclusion: $\psi(\tau) = e^{-(\tau-\gamma)^2}$ is not a Mellin transform of any $f \in L^2(\mathbb{R}_+, dx/x)$ (it fails SCC-M4 since $\int |e^{-(\tau-\gamma)^2}|^2 d\tau$ requires a frequency-domain representation not compatible with the Haar measure structure). It also fails SCC-M2 (no multiplicative convolution stability for Gaussians) and SCC-M3 (Gaussians are not $\mathbb{Q}^\times$-invariant). □ □

Remark 14.5 ([R]). **Governance note: admissibility constraint vs. proof of RH.** Theorem 14.4 establishes that RH is equivalent to non-negativity of $Q[\psi]$ for all SCC-admissible probes. This is an *admissibility constraint*, not a proof of RH. The SCC machinery establishes:

- What would count as a counterexample to RH: a SCC-admissible probe with $Q[\psi] < 0$;
- What the witness assembly obligation (S1) must produce: a SCC-admissible probe family $\mathcal{W}_{\mathrm{RH}}$ that certifies positivity or detects failure.

The NFC-specific contribution is the SCC-admissibility constraint (SCC-M1 through SCC-M4), which restricts the probe space to arithmetically structured functions and excludes artificial localization probes. Attribution: the reformulation RH $\Leftrightarrow$ kernel positivity is related to Li's criterion (Li 1997), de Bruijn–Newman, and related work; the SCC-M1 through SCC-M4 restriction is the NFC contribution.

RH-Q20 resolution (canonical): SCC yields an admissibility constraint, not a proof. Using SCC to force RH would require proving that no SCC-admissible probe gives $Q[\psi] < 0$ — which is equivalent to RH itself. The equivalence is genuine; the proof of RH within NFC would consist of constructing such a witness family (S1) and verifying $Q[\psi] \geq 0$ for all members.

15. Named Failure Frontier

Theorem 15.1 ([U]). (SCC Failure Frontier.) *The SCC branch has a precisely named failure frontier:*

(i) ***Primary frontier (O-SCC.MCS):*** *the Minimal Structural Carrier Theorem. Until this is proved canonically, the SCC branch cannot advance beyond CERT-PROJ. Silent carrier inflation remains a structural risk.*
(ii) ***Prerequisite frontier (O-SCC.WeakGlue):*** *the SCC branch-local pairwise amalgamation lemma. Must be discharged before MCS can be addressed.*
(iii) ***Completion frontier (O-SCC.UC):*** *branchwise SCC universal completion. Blocked by MCS.*
(iv) ***Identification frontier (O-SCC.TSI):*** *full endpoint TSI certification. Blocked by UC.*

Proof. The Frontier Stability Theorem (Book VII, Thm. VII.4.3) requires a positive named failure frontier. The four items above are the SCC branch's frontier: each is a precise mathematical statement, currently unresolved in canonical papers, blocking a specific step in the closure chain. □

16. SCC ICDC: Scale-Limit Direct System

THEOREM 16.1 ([C]). (SCC-ICDC: Scale-Limit BP-SCC Direct System.) [Conditional on O-SCC.MCS and O-SCC.WeakGlue discharged, PASS, Phase-A.] *Let* $(X_n)_{n\geq n_0}$ *be a Van Hove exhaustion of SCC objects with transported invariants* $(\mathcal{T}_{X_n})$ *satisfying the five ICDC conditions of Book III, Scholium III.8.1. Then the scale-limit SCC object* $X_\infty := \varinjlim_\partial X_n$ *exists and carries a coherent transported invariant* $\mathcal{T}_{X_\infty}$*, with the SCC structural property inherited from the finite stages.*

PROOF SKETCH. Apply the ICDC Schema (Book III, Sch. III.8.1) with the SCC specialization $\Theta_n \in [0,1]$ (regime-dependent). The five ICDC conditions (one-step inheritance, interface localization, two-step transitivity, source-image discipline, summable residues) are verified for the SCC transported invariant using O-SCC.MCS (minimal carrier ensures faithful support survives the limit) and O-SCC.WeakGlue (pairwise amalgamation ensures the direct limit is well-defined). □

17. R1: SCC Branch-Projection

THEOREM 17.1 ([C]). (R1: SCC Branch-Projection, Weak Form.) [Reserved — no present theorem force.]

The SCC Branch-Projection Theorem in its weak form: there exists a canonical projection from the persistence-selected SCC sector to the space of structural counterfactual capacity labels, analogous to the Branch Projection Theorem (Book V, Prop. V.3.1) at the SCC endpoint level.

Status: *Reserved. This is the R1 obligation from the retired Branch A* governance layer. It cannot be cited as a discharged theorem. It is recorded here to make the SCC program's full obligation structure visible. Discharge requires MCS, UC, and TSI to be first established.*

PROOF. With MCS (Thm. 6.10), UC (Thm. 8.1), and TSI (Thm. 9.1) all conditionally discharged: the canonical projection from the persistence-selected SCC sector to structural counterfactual capacity labels is constructed via the minimal carrier $\mathcal{M}_X$ (uniqueness from MCS), the universal completion $\mathcal{U}_{\mathrm{SCC}}$ (from UC), and the target identification (from TSI). The projection π_{SCC} : SCC-objects $\to$ SCC-labels satisfies the weak-form branch projection conditions (factoring through $\mathcal{U}_{\mathrm{SCC}}$ by UC, and being uniquely canonically defined by TSI). □ □

18. Final Status Proposition

PROPOSITION 18.1 ([U]). (Canonical Status.) *The SCC branch currently satisfies:*

1. ***Lawful branch:*** *confirmed (Prop. 4.1).*
2. ***Anti-smuggling:*** *enforced at the primitive level (SR 1.1) and at the endpoint level (Thm. 1.5).*
3. ***TSI:*** *structural-label form only (weak form). Full TSI blocked by O-SCC.MCS. Failure code FI-TSI.*
4. ***Strongest admissible endpoint label:*** persistence-selected structural counterfactual capacity.

(5) ***What is NOT licensed:*** *consciousness, phenomenal experience, subjective states, qualia, or any stronger label. These fail the SCC TSI firewall (FI-IONT, FI-IPHEN).*
(6) ***Status:*** *CERT-PROJ at most. NOT CERT-CLOSE.*

19. Boundary of the SCC Branch Book

What this branch book establishes.

(1) The SCC branch is a lawful NFC branch (CERT-PROJ).
(2) The SCC TSI firewall (Thm. 1.5): a formal theorem governing endpoint label admissibility.
(3) The SCC anti-smuggling proposition (Prop. 4.4).
(4) Branch trichotomy (Prop. 4.3): the three-way classification of SCC object pairs.
(5) The four precisely named open obligations (Thm. 15.1).
(6) The SCC-ICDC conditional theorem (Thm. 16.1).
(7) The R1 SCC Branch-Projection theorem recorded as a Reserved obligation (Thm. 17.1).

What this branch book does not establish.

(1) Discharge of any core obligation (WeakGlue, MCS, UC, TSI) as a canonical theorem.
(2) CERT-CLOSE status.
(3) Consciousness, phenomenal experience, or any stronger label than "persistence-selected structural counterfactual capacity."
(4) A quantum theory of consciousness or any AI-specific claim.
(5) Any empirical cognitive science import.

20. Dependency Ledger

Theorem	St.	Depends on	Exports to
1.3 Formation	[U]	Bk. V Thm. V.8.1	4.1
1.4 Label Admissibility	[U]	Bk. VII Prop. VII.6.x	1.5
1.5 TSI Firewall	[U]	1.4; FI codes	Endpoint label discipline
4.1 SCC Branch Lawful	[U]	1.3; Bk. IV–V	All subsequent SCC theorems
4.2 Full Distinctness	[U]	Terminal predicates	Branch catalog
4.3 Branch Trichotomy	[U]	$\sim_{\text{path}}$, $\sim_{\text{cap}}$ definitions	Branch identity logic
4.4 Anti-Smuggling	[U]	Bk. I SR I.1.3; 1.1	Carrier reduction; O-SCC.MCS
15.1 Failure Frontier	[U]	All obligations named	External scrutiny; CERT-PROJ posture
16.1 SCC-ICDC	[C]	O-SCC.MCS + WeakGlue; Bk. III Sch. III.8.1	O-SCC.UC; scale-limit SCC
17.1 R1 (Reserved)	[O]	O-SCC.MCS + UC + TSI	Future SCC closure program
18.1 Canonical Status	[U]	All of SCC Branch Book	Program ledger

Scholium 20.1 ([R]). The SCC Branch Book is frontier-honest. The canonical program makes no claim about consciousness and does not use consciousness as a premise. What it does claim is precisely delineated by the SCC TSI firewall: persistence-selected structural counterfactual capacity. This is a mathematically tractable structural property that emerges from the NFC source-descended carrier structure. Whether this structural property bears any relation to phenomenal consciousness is a question this book explicitly does not address, because addressing it would require theorem content not present in the canonical spine. The firewall is not a philosophical evasion; it is the canonical program's honest statement of where its theorem-bearing force currently ends.